THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 2: **Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If α, β are zeroes of $x^2 - 7x + 6$, then $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is: 1

(a) 37 **(b)** $\frac{7}{6}$ **(c)** $\frac{49}{6}$ **(d)** $\frac{37}{6}$

2. For the cubic with zeroes $2, 3, 4$, $\alpha\beta + \beta\gamma + \gamma\alpha$ is: 1

(a) 24 **(b)** 9 **(c)** -26 **(d)** 26

3. If the zeroes of $x^2 - 12x + k$ differ by 2, then k is: 1

(a) 35 **(b)** 37 **(c)** 34 **(d)** 70

4. If $x = 3$ is a zero of $x^2 - 11x + 24$, the other zero is: 1

(a) 8 **(b)** 14 **(c)** 21 **(d)** -8

5. If α, β are zeroes of $x^2 - 4x - 5$, then $\alpha^3 + \beta^3$ is: 1

(a) 64 **(b)** 124 **(c)** 4 **(d)** 104

6. If α, β are zeroes of $x^2 - 8x + 15$, then $\alpha^2\beta + \alpha\beta^2$ is: 1

(a) 23 **(b)** 225 **(c)** 120 **(d)** 64

7. If α, β are zeroes of $x^2 - x - 12$, then $|\alpha - \beta|$ is: 1

(a) 1 **(b)** 14 **(c)** 8 **(d)** 7

8. If α, β are zeroes of $x^2 - 9x + 18$, then $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is: 1

(a) 45 **(b)** $\frac{5}{2}$ **(c)** $\frac{9}{2}$ **(d)** $\frac{1}{2}$

9. **Assertion (A):** The polynomial $x^2 + 1$ has two real zeros. 1

Reason (R): A quadratic polynomial can have at most two zeros.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The degree of the polynomial $5x^3 - x + 7$ is 3. 1

Reason (R): Every linear polynomial has exactly one zero.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. If the sum of the zeroes of $x^2 - 9x + k$ is equal to their product, find k. 2

12. If one zero of $x^2 - 10x + k$ is 4 times the other, find k. 2

13. If α, β are zeroes of $x^2 - 6x + 5$, form a quadratic polynomial whose zeroes are α^2 and β^2. 2

14. If the sum of the zeroes of $x^2 - 11x + k$ is equal to their product, find k. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find a quadratic polynomial whose zeroes are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$, where α, β are zeroes of $x^2 - 5x + 4$. 3

16. The zeroes of $x^2 - 14x + c$ are α and β with $\alpha - \beta = 8$. Find c. 3

17. If α, β, γ are the zeroes of $x^3 - 9x^2 + 26x - 24$, find $\alpha^2 + \beta^2 + \gamma^2$. 3

OR

Find a quadratic polynomial whose zeroes are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$, where α, β are zeroes of $x^2 - 7x + 12$. 3

18. The zeroes of $x^2 - 16x + c$ are α and β with $\alpha - \beta = 10$. Find c. 3

OR

If α, β, γ are the zeroes of $x^3 - 7x^2 + 7x + 15$, find $\alpha^2 + \beta^2 + \gamma^2$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Sunita's rectangular vegetable plot has length and breadth (in metres) that are the zeros of the polynomial $x^2 - 30x + 216$. 5

(i) Find the sum of the length and breadth.

(ii) Find the length and the breadth by factorising.

(iii) Without finding the zeros, state the area of the plot.

(iv) Find the perimeter of the plot and the cost of fencing it at ₹ 50 per metre.

20. The zeroes of $x^3 - 3x^2 - x + 3$ are in arithmetic progression. Find them. 5

OR

If α, β are the zeroes of $x^2 - 2x - 24$, form a quadratic polynomial whose zeroes are $\alpha + \beta$ and $\alpha\beta$. 5

END OF QUESTION PAPER — Total: 40 Marks